

Conjecture 00000001463: ℓ_∞^n does not attain $c\sqrt{n}$

Conjecture and reading

The supremum of the projection constants of ℓ_p^n over $p \in [1, \infty]$ is $c\sqrt{n}$ uniformly (explicit c), attained asymptotically by both ℓ_1^n and ℓ_∞^n .

Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and let $\ell_p^n = \mathbb{K}^n$ with the p -norm (Mathlib PiLp p). For a finite-dimensional normed space E (all spaces in the conjecture are of this kind), an isometric embedding $j : E \rightarrow X$ into a normed space X , put

$$\lambda(E, j) = \inf\{\|P\| : P : X \rightarrow X \text{ bounded linear, } P(jy) = jy \ \forall y, \ P(X) \subseteq j(E)\},$$

and the *absolute projection constant* $\lambda(E) = \sup_{X, j} \lambda(E, j)$ over all Banach spaces X (of the base universe) and isometric embeddings j . For finite-dimensional E these real-valued infima and suprema have their intended meaning: Hahn–Banach extensions of coordinate functionals give admissible projections in every superspace, so each infimum is over a nonempty set, and an Auerbach basis bounds them by $\dim E$, so the supremum is over a bounded set. (For infinite-dimensional E a copy need not be complemented, and the real-valued definitions would need existence and boundedness hypotheses; they are not used here.) This is the definition formalized in Lean (`relProj`, `absProj`, `lam`).

ConjSharp: there is a real c with $\sup_p \lambda(\ell_p^n) = c\sqrt{n}$ for all $n \geq 1$, $\lambda(\ell_1^n)/(c\sqrt{n}) \rightarrow 1$ and $\lambda(\ell_\infty^n)/(c\sqrt{n}) \rightarrow 1$ (any “explicit formula” for c is covered, since c is quantified over all reals). **ConjOrder** (weaker reading of “attained asymptotically”): $\exists \kappa > 0, \forall n \geq 1, \kappa\sqrt{n} \leq \lambda(\ell_\infty^n)$.

Theorem

For $\mathbb{K} = \mathbb{R}$ and $\mathbb{K} = \mathbb{C}$, neither **ConjSharp** nor **ConjOrder** holds. (Lean: `C1463.main : C1463.Claim`.)

Proof

Lemma 1. *For every n , every Banach (indeed normed) space X and every isometric embedding $j : \ell_\infty^n \rightarrow X$, $\lambda(\ell_\infty^n, j) \leq 1$. (Lean: `relProj_le_one`.)*

Proof. The coordinate functional $y \mapsto y_i$ transported to $j(\ell_\infty^n)$ through the isometry j has norm ≤ 1 . By Hahn–Banach (Mathlib `exists_extension_norm_eq`) it extends to $g_i \in X^*$ with $\|g_i\| \leq 1$. Put $Tx = (g_i x)_i$; then $\|Tx\|_\infty \leq \|x\|$ and $T(jy) = y$. The map $P = j \circ T$ satisfies $P(jy) = jy$, has range in $j(\ell_\infty^n)$ and $\|P\| \leq 1$. $\square$

Hence $0 \leq \lambda(\ell_\infty^n) \leq 1$ (the lower bound because every $\lambda(E, j)$ is an infimum of norms; Lean: `lam_infty_le_one`, `lam_infty_nonneg`). Write $a_n = \lambda(\ell_\infty^n) \in [0, 1]$.

ConjSharp fails. Suppose $a_n/(c\sqrt{n}) \rightarrow 1$. If $c < 0$ the quotients are ≤ 0 for $n \geq 1$; if $c = 0$ they are 0; neither tends to 1. If $c > 0$ then $a_n/(c\sqrt{n}) \leq 1/(c\sqrt{n}) \rightarrow 0 \neq 1$. (Lean: `not_sharp`.)

ConjOrder fails. If $\kappa\sqrt{n} \leq a_n \leq 1$ for all $n \geq 1$, then $\kappa^2 n \leq 1$ for all $n \geq 1$, false for $n > 1/\kappa^2$. (Lean: `not_order`.)

Scope and conventions

Refuted: both readings above, for $\mathbb{R}$ and $\mathbb{C}$, for every real c . Not addressed: relative projection constants in a fixed superspace, and the claims about ℓ_1^n and the exact supremum (not needed). λ is the real-valued `sSup` of a set of reals; only $0 \leq \lambda(\ell_\infty^n) \leq 1$ is used, so no junk-value convention matters. In fact $\lambda(\ell_\infty^n) = 1$ since $n \geq 1$.

Lean audit

File `lean/Conjecture1463/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1); main theorem `C1463.main` : `C1463.Claim`, `Claim = ¬ConjSharp ℝ ∧ ¬ConjSharp ℂ ∧ ¬ConjOrder ℝ ∧ ¬ConjOrder ℂ`. Axioms: `propext`, `Classical.choice`, `Quot.sound`; no `sorry`, no `native_decide`. No external literature is cited.